

(1) Base solution with output:

```
package base.client;

import base.core.PaymentService;
import base.external.LegacyPayService;
import base.integration.LegacyPayIntegration;

public class Main {
    public static void main(String[] args) {
        LegacyPayService legacyService = new LegacyPayService();
        PaymentService service = new LegacyPayIntegration(legacyService);
        PaymentClient client = new PaymentClient(service);
        client.makePayment(250.0);
    }
}
```

```
package base.client;

import base.core.PaymentService;

public class PaymentClient {

    private PaymentService paymentService;

    public PaymentClient(PaymentService paymentService) {
        this.paymentService = paymentService;
    }

    public void makePayment(double amount) {
        System.out.println("Client initiating payment...");
        paymentService.pay(amount);
        System.out.println("Payment completed successfully.");
    }
}
```

```
}
```

```
package base.core;

public interface PaymentService {
    void pay(double amount);
}
```

```
package base.external;

public class LegacyPayService {
    public void makePayment(double valueInDollars) {
        System.out.println("[LegacyPayService] Processing payment of $" + valueInDollars);
    }
}
```

```
package base.integration;

import base.core.PaymentService;
import base.external.LegacyPayService;

public class LegacyPayIntegration implements PaymentService {

    private LegacyPayService legacyPayService;

    public LegacyPayIntegration(LegacyPayService legacyPayService) {
        this.legacyPayService = legacyPayService;
    }

    @Override
    public void pay(double amount) {
        legacyPayService.makePayment(amount);
    }
}
```

```
PS F:\Asst.Prof.Engr.A.Rahman\SE-Sp2026> & 'F:\jdk-21.0.5\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'F:\VSCode-win32-x64-1.98.2\data\user-data\User\workspaceStorage\07d17c5476f9307329cc72a5f308c93c\redhat.java\jdt_ws\SE-Sp2026_c5b18d9e\bin' 'base.client.Main'  
Client initiating payment...  
[LegacyPayService] Processing payment of $250.0  
Payment completed successfully.  
PS F:\Asst.Prof.Engr.A.Rahman\SE-Sp2026>
```

Exercise 1 solution with output:

```
package exercise1.client;

import base.client.PaymentClient;
import base.core.PaymentService;
import exercise1.external.CentPayService;
import exercise1.integration.CentPayIntegration;

public class Main {
    public static void main(String[] args) {
        PaymentService service = new CentPayIntegration(new CentPayService());
        PaymentClient client = new PaymentClient(service);
        client.makePayment(99.99);
    }
}
```

```
package exercise1.external;

public class CentPayService {
    public void processPayment(int amountInCents) {
        System.out.println("[CentPayService] Processing payment of " + amountInCents + " cents");
    }
}
```

```
package exercise1.integration;

import base.core.PaymentService;
import exercise1.external.CentPayService;

public class CentPayIntegration implements PaymentService {

    private CentPayService centPayService;
```

```
public CentPayIntegration(CentPayService centPayService) {  
    this.centPayService = centPayService;  
}  
  
@Override  
public void pay(double amount) {  
    int cents = (int)(amount * 100);  
    centPayService.processPayment(cents);  
}  
}
```

```
PS F:\Asst.Prof.Engr.A.Rahman\SE-Sp2026> f::; cd 'f:\Asst.Prof.Engr.A.Rahman\SE-Sp2026'; & 'F:\jdk-21.0.5\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'F:\VSCode-win32-x64-1.98.2\data\user-data\User\workspaceStorage\07d17c5476f9307329cc72a5f308c93c\redhat.java\jdt_ws\SE-Sp2026_c5b18d9e\bin' 'exercise1.client.Main'  
Client initiating payment...  
[CentPayService] Processing payment of 9999 cents  
Payment completed successfully.  
PS F:\Asst.Prof.Engr.A.Rahman\SE-Sp2026> 
```

Exercise 2 solution with output:

```
package exercise2.client;

import base.client.PaymentClient;
import base.core.PaymentService;
import exercise2.external.GlobalPayService;
import exercise2.integration.GlobalPayIntegration;

public class Main {
    public static void main(String[] args) {
        PaymentService service = new GlobalPayIntegration(new GlobalPayService(), "USD");
        PaymentClient client = new PaymentClient(service);
        client.makePayment(150.75);
    }
}
```

```
package exercise2.external;

public class GlobalPayService {
    public void executeTransaction(String currencyCode, double value) {
        System.out.println("[GlobalPayService] Processing " + value + " " + currencyCode);
    }
}
```

```
package exercise2.integration;

import base.core.PaymentService;
import exercise2.external.GlobalPayService;

public class GlobalPayIntegration implements PaymentService {

    private GlobalPayService service;
    private String currency;
```

```
public GlobalPayIntegration(GlobalPayService service, String currency) {  
    this.service = service;  
    this.currency = currency;  
}  
  
@Override  
public void pay(double amount) {  
    service.executeTransaction(currency, amount);  
}  
}
```

```
PS F:\Asst.Prof.Engr.A.Rahman\SE-Sp2026> f.; cd 'f:\Asst.Prof.Engr.A.Rahman\SE-Sp2026'; & 'F:\jdk-21.0.5\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'F:\VSCode-win32-x64-1.98.2\data\user-data\User\workspaceStorage\07d17c5476f9307329cc72a5f308c93c\redhat.java\jdt_ws\SE-Sp2026_c5b18d9e\bin' 'exercise2.client.Main'  
Client initiating payment...  
[GlobalPayService] Processing 150.75 USD  
Payment completed successfully.  
PS F:\Asst.Prof.Engr.A.Rahman\SE-Sp2026> 
```

Exercise 3 solution with output:

```
package exercise3.client;

import base.client.PaymentClient;
import base.core.PaymentService;
import exercise3.external.ConfirmPayService;
import exercise3.integration.ConfirmPayIntegration;

public class Main {
    public static void main(String[] args) {
        PaymentService service = new ConfirmPayIntegration(new ConfirmPayService());
        PaymentClient client = new PaymentClient(service);
        client.makePayment(300.0);
    }
}
```

```
package exercise3.external;

public class ConfirmPayService {
    public String sendPayment(double amount) {
        return "TXN-OK :: Amount=" + amount;
    }
}
```

```
package exercise3.integration;

import base.core.PaymentService;
import exercise3.external.ConfirmPayService;

public class ConfirmPayIntegration implements PaymentService {

    private ConfirmPayService service;
```



```
public ConfirmPayIntegration(ConfirmPayService service) {  
    this.service = service;  
}  
  
@Override  
public void pay(double amount) {  
    String confirmation = service.sendPayment(amount);  
    System.out.println("[AlphaSoft Log] Payment successful -> " + confirmation);  
}  
}
```

```
PS F:\Asst.Prof.Engr.A.Rahman\SE-Sp2026> f:: cd 'f:\Asst.Prof.Engr.A.Rahman\SE-Sp2026'; & 'F:\jdk-21.0.5\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'F:\VSCode-win32-x64-1.98.2\data\user-data\User\workspaceStorage\07d17c5476f9307329cc72a5f308c93c\redhat.java\jdt_ws\SE-Sp2026_c5b18d9e\bin' 'exercise3.client.Main'  
Client initiating payment...  
[AlphaSoft Log] Payment successful -> TXN-OK :: Amount=300.0  
Payment completed successfully.  
PS F:\Asst.Prof.Engr.A.Rahman\SE-Sp2026> []
```

Exercise 4 solution with output:

```
package exercise4.client;

import base.client.PaymentClient;
import base.core.PaymentService;
import base.external.LegacyPayService;
import base.integration.LegacyPayIntegration;
import exercise2.external.GlobalPayService;
import exercise2.integration.GlobalPayIntegration;
import exercise4.integration.PaymentProviderSelector;

public class Main {
    public static void main(String[] args) {
        PaymentService legacy = new LegacyPayIntegration(new LegacyPayService());
        PaymentService global = new GlobalPayIntegration(new GlobalPayService(), "EUR");

        PaymentProviderSelector selector = new PaymentProviderSelector(legacy);
        PaymentClient client = new PaymentClient(selector);

        client.makePayment(120.0);
        selector.switchProvider(global);
        client.makePayment(120.0);
    }
}
```

```
package exercise4.integration;

import base.core.PaymentService;

public class PaymentProviderSelector implements PaymentService {

    private PaymentService provider;
```

```

public PaymentProviderSelector(PaymentService provider) {
    this.provider = provider;
}

public void switchProvider(PaymentService newProvider) {
    this.provider = newProvider;
    System.out.println("[System] Payment provider switched.");
}

@Override
public void pay(double amount) {
    provider.pay(amount);
}
}

```

```

PS F:\Asst.Prof.Engr.A.Rahman\SE-Sp2026> f::; cd 'f:\Asst.Prof.Engr.A.Rahman\SE-Sp2026'; & 'F:\jdk-21.0.5\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'F:\VSCode-win32-x64-1.98.2\data\user-data\User\workspaceStorage\07d17c5476f9307329cc72a5f308c93c\redhat.java\jdt_ws\SE-Sp2026_c5b18d9e\bin' 'exercise4.client.Main'
Client initiating payment...
[LegacyPayService] Processing payment of $120.0
Payment completed successfully.
[System] Payment provider switched.
Client initiating payment...
[GlobalPayService] Processing 120.0 EUR
Payment completed successfully.
PS F:\Asst.Prof.Engr.A.Rahman\SE-Sp2026> 

```

Exercise 5 solution with output:

```
package exercise5.client;

import base.client.PaymentClient;
import base.core.PaymentService;
import exercise5.external.MockPayService;
import exercise5.integration.MockPayIntegration;

public class Main {
    public static void main(String[] args) {
        PaymentService service = new MockPayIntegration(new MockPayService());
        PaymentClient client = new PaymentClient(service);
        client.makePayment(500.0);
    }
}
```

```
package exercise5.external;

public class MockPayService {
    public void simulate(double amount) {
        System.out.println("[MockPayService] Simulating payment of $" + amount);
    }
}
```

```
package exercise5.integration;

import base.core.PaymentService;
import exercise5.external.MockPayService;

public class MockPayIntegration implements PaymentService {

    private MockPayService service;
```

```
public MockPayIntegration(MockPayService service) {
    this.service = service;
}

@Override
public void pay(double amount) {
    service.simulate(amount);
    System.out.println("[Test Mode] No real transaction executed.");
}
}
```

```
PS F:\Asst.Prof.Engr.A.Rahman\SE-Sp2026> f.; cd 'f:\Asst.Prof.Engr.A.Rahman\SE-Sp2026'; & 'F:\jdk-21.0.5\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'F:\VSCode-win32-x64-1.98.2\data\user-data\User\workspaceStorage\07d17c5476f9307329cc72a5f308c93c\redhat.java\jdt_ws\SE-Sp2026_c5b18d9e\bin' 'exercise5.client.Main'
Client initiating payment...
[MockPayService] Simulating payment of $500.0
[Test Mode] No real transaction executed.
Payment completed successfully.
PS F:\Asst.Prof.Engr.A.Rahman\SE-Sp2026> █
```